Callable Bonds under the Hull–White Model: European, Bermudan and American Calls with Three Numerical Methods

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Abstract

We price fixed-rate callable bonds with issuer default risk under a one-factor Hull–White short rate fitted to the euro area AAA government curve of 5 October 2026 and calibrated to co-terminal swaptions ($a = 0.0863$, $\sigma = 111.9$bp, RMSE 0.76bp of normal volatility), with a deterministic default intensity bootstrapped from CDS spreads. The European, Bermudan and American calls differ only by the set of dates where the issuer's choice $V \leftarrow \min(V, K + \mathrm{AI})$ is applied, and three independent methods solve the same problem: a Crank–Nicolson PDE, a Hull–White trinomial tree with unequal time steps, and Monte Carlo with a Longstaff–Schwartz call policy estimated and applied on independent paths. On a seasoned 4.75% bond maturing in 2036 and callable at par from 2029, the three methods agree within 0.2 cent per 100 (tree) and 1.5 standard errors (Monte Carlo); the PDE reprices the straight bond to 3×10^{-7} and the European callable bond to 10^{-4} of Jamshidian's closed form. The issuer's option is worth 1.20, 1.74 and 1.85 per 100 for the European, Bermudan and American calls (6.74 to 7.40 for the same bond without default), the OAS at a clean price of 98.50 falls from 139bp (Z-spread) to 106, 93 and 90bp, and the effective convexity turns from +67 to between −70 and −92. At identical calibration to European swaptions, the mean reversion moves the American price by up to 0.44 per 100 and the European by 0.01: the value of the right to choose the call date is model risk.

Contents

1 Introduction **3**

2 Model and Product **3**
2.1 Hull–White short rate . . . 3
2.2 Issuer default . . . 3
2.3 The product . . . 4
2.4 Pricing problem . . . 4

3 Numerical Methods **4**
3.1 Crank–Nicolson PDE . . . 4
3.2 Trinomial tree . . . 5
3.3 Monte Carlo and Longstaff–Schwartz . . . 5
3.4 Closed forms used as benchmarks . . . 5
3.5 Spreads, yields and risk . . . 6

4 Market Data and Calibration **6**

5 Validation **7**
5.1 Closed forms . . . 7
5.2 Agreement of the three methods . . . 7
5.3 Longstaff–Schwartz bias . . . 7
5.4 From Bermudan to American . . . 8
5.5 Grid convergence . . . 8

6 Prices, Spreads and Risk **8**
6.1 Prices . . . 8
6.2 Spreads and yields from a quoted price . . . 9
6.3 Interest rate, volatility and credit risk . . . 9

7 Exercise Policy **10**
7.1 Exercise boundary . . . 10
7.2 Call probabilities and expected life . . . 11

8 Model Risk and Term Sheet Variants **12**
8.1 Mean reversion . . . 12
8.2 Volatility and credit . . . 12
8.3 Term sheet variants . . . 12

9 Implementation **13**

10 Limitations and Next Steps **14**

11 Conclusion **14**

A Notation **15**

1 Introduction

A callable bond is a fixed-rate bond that its issuer may redeem before maturity at a contractual call price plus accrued interest. Issuers call when rates, or their own credit spread, have fallen enough to refinance more cheaply; the bondholder is short this option and is compensated by a higher coupon. From the holder's side,

$$\text{callable bond} = \text{straight bond} - \text{issuer call option}. \tag{1}$$

Three exercise styles are common: *European* (one call date, at the end of the call protection), *Bermudan* (every coupon date after the protection, the most frequent structure for corporate and agency bonds) and *American* (any time in the call period, typical of US corporate bonds with make-whole or par calls). The call is an option on a coupon bond, hence on rates: a short-rate model fitted to the yield curve is the natural framework [6, 3, 2], and the one-factor Hull–White model is the market standard for this product because zero-coupon bonds and their options are in closed form.

This report documents a complete pricer: from a term sheet in calendar dates to prices, spreads, yields, risk measures and the exercise policy, for the three styles. Its distinctive feature is that every number is obtained by three independent methods (PDE, tree, Monte Carlo) and checked against closed forms where they exist. Section 2 sets out the model and the product, Section 3 the three numerical methods, Section 5 the validation, Section 6 the prices, spreads and risk of an illustrative bond, Section 7 the exercise policy and Section 8 the model risk.

2 Model and Product

2.1 Hull–White short rate

Under the risk-neutral measure $\mathbb{Q}$, the short rate follows [6]

$$\mathrm{d}r_t = \big(\theta(t) - a r_t\big)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \tag{2}$$

written $r_t = x_t + \alpha(t)$ with the Ornstein–Uhlenbeck state $\mathrm{d}x_t = -a x_t\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$, $x_0 = 0$, and the deterministic shift

$$\alpha(t) = f(0,t) + \frac{\sigma^2}{2a^2}\big(1 - e^{-at}\big)^2, \tag{3}$$

which makes the model reprice every discount factor $P(0,T)$ of the initial curve, $f(0,t)$ being the instantaneous forward rate [3]. Zero-coupon bonds are exponential-affine in the state:

$$\begin{aligned} P(t,T;x) &= \frac{P(0,T)}{P(0,t)}\exp\big(-B(t,T)\,x - \Gamma(t,T)\big), \qquad B(t,T) = \frac{1 - e^{-a(T-t)}}{a}, \\ \Gamma(t,T) &= \frac{\sigma^2}{2a^2}B(t,T)\big(1 - e^{-at}\big)^2 + \frac{\sigma^2}{4a}\big(1 - e^{-2at}\big)B(t,T)^2. \end{aligned} \tag{4}$$

A higher state lowers every bond price: the issuer calls when x, hence r, is low.

2.2 Issuer default

Default is the first jump of a Cox process with deterministic, piecewise constant intensity $\lambda(t)$, independent of rates [10, 4]; the survival probability is $Q(0,t) = \exp(-\int_0^t \lambda)$. At default the holder receives RN (recovery of par). The intensities are bootstrapped sequentially from CDS par spreads (quarterly premiums with accrual on default), each one repricing its CDS exactly. With $\lambda \equiv 0$ the bond is default-free (government or agency).

2.3 The product

The bond pays coupons c_i at dates $T_1 < \cdots < T_n$ and the notional N at T_n. The issuer may redeem it on the call set $\mathcal{T}$ at the clean call price $K(t)$ plus accrued interest $\mathrm{AI}(t)$; on a coupon date the holder receives the coupon and the call price. The three styles are

$$\mathcal{T}^{\mathrm{Eur}} = \{T_c\}, \qquad \mathcal{T}^{\mathrm{Ber}} = \{T_i : T_c \le T_i < T_n\}, \qquad \mathcal{T}^{\mathrm{Am}} = [T_c, T_n], \tag{5}$$

with T_c the end of the call protection. The term sheet is given in calendar dates: model time is ACT/365F from the valuation date, coupon dates roll back from the maturity date, each regular coupon pays Nc/f (ACT/ACT ICMA) and $\mathrm{AI}(t)$ is the running coupon pro rata of its period. A seasoned bond carries accrued interest at the valuation date; the model gives the *dirty* price, and the market quotes the *clean* price, dirty minus AI(0).

2.4 Pricing problem

The issuer minimises the value of its debt. Before default, the value $V(t,x)$ of the bond is the value of an optimal stopping problem, $V(0,0) = \inf_{\tau\in\mathcal{T}} \mathbb{E}\big[\text{discounted flows until } \tau\big]$, and by Feynman–Kac it solves, between payment and call dates,

$$\partial_t V - ax\,\partial_x V + \tfrac{1}{2}\sigma^2\,\partial_{xx} V - \big(x + \alpha(t) + \lambda(t)\big)V + \lambda(t) RN = 0, \tag{6}$$

with $V(T_n^-, x) = N + c_n$ and, at a date t,

$$V(t^-, x) = \min\big(V(t^+, x),\ K(t) + \mathrm{AI}(t)\big)\,\mathbf{1}_{t\in\mathcal{T}} + V(t^+, x)\,\mathbf{1}_{t\notin\mathcal{T}} + c(t), \tag{7}$$

where $c(t)$ is the coupon paid at t, if any. Default enters as an extra discount rate λ and a recovery source term. The three styles differ only by $\mathcal{T}$: one solver serves all of them. On a call date, the *critical state* $x^*(t) = \max\{x : V(t^+, x) \ge K(t) + \mathrm{AI}(t)\}$ separates calling (below) from continuing, and $r^*(t) = x^*(t) + \alpha(t)$ is the critical short rate.

3 Numerical Methods

The three methods share the time grid: the valuation date, every coupon, call and CDS pillar date, and between two of them equal steps of at most 1/100 year (975 steps for the bond of Section 6). On the American call period, the issuer may call at every node.

3.1 Crank–Nicolson PDE

Since α and λ do not depend on x, on each step $[t_{k-1}, t_k]$ the solution is

$$V(t_{k-1}, x) = e^{-\int_{t_{k-1}}^{t_k}\alpha}\,\frac{Q(0,t_k)}{Q(0,t_{k-1})}\,U(t_{k-1}, x) + RN\Big(1 - \frac{Q(0,t_k)}{Q(0,t_{k-1})}\Big)P\big(t_{k-1}, t_{k-1} + \tfrac{\Delta t}{2}; x\big), \tag{8}$$

where U solves the time-homogeneous equation $U_t - axU_x + \frac{1}{2}\sigma^2 U_{xx} - xU = 0$ with $U(t_k) = V(t_k)$, and the recovery is paid at mid-step. The deterministic discounting $\int\alpha$ is exact (closed form), and the operator of U is constant, so each step is one tridiagonal solve of the θ-scheme

$$(I - \theta\,\Delta t\,L)\,U^{k-1} = (I + (1-\theta)\,\Delta t\,L)\,U^k, \tag{9}$$

with L the central-difference discretisation of $-ax\partial_x + \frac{1}{2}\sigma^2\partial_{xx} - x$, one-sided and upwind at the boundaries, where the drift points inward. The grid is uniform in x over ± 6 standard deviations of $x(T_n)$ with 401 nodes. $\theta = \frac{1}{2}$ (Crank–Nicolson) except for two fully implicit steps after each discrete call date (Rannacher smoothing [12]), which damp the oscillations created by the kink of the min in (7). The PDE returns $V(0,x)$ for every initial state, hence the price as a function of $r(0)$, and the critical states $x^*(t)$.

3.2 Trinomial tree

The Hull–White tree [7] is built on the state x and on the same unequal time grid. Level $k+1$ has its own spacing, set by the variance of the step that leads to it,

$$\Delta x_{k+1} = \sqrt{3V_k}, \qquad V_k = \sigma^2 \frac{1 - e^{-2ah_k}}{2a}, \qquad h_k = t_{k+1} - t_k. \tag{10}$$

A node x of level k branches to the node $j\,\Delta x_{k+1}$ nearest to its conditional mean xe^{-ah_k} and to its two neighbours. With $\eta = (xe^{-ah_k} - j\Delta x_{k+1})/\Delta x_{k+1} \in [-\frac{1}{2}, \frac{1}{2}]$ and $m = \frac{1}{3} + \eta^2$, the probabilities matching the mean and variance are

$$p_u = \frac{m+\eta}{2}, \qquad p_m = 1 - m, \qquad p_d = \frac{m-\eta}{2}, \tag{11}$$

all positive ($p_m \geq \frac{5}{12}$, $p_u, p_d \geq \frac{1}{24}$) whatever the step. Levels are truncated at ± 7 standard deviations of $x(t)$ (179 nodes at maturity). The shift is not taken from the closed form: the discount factor of a node over a step is $e^{-(\alpha_k + x)h_k}$ and α_k is solved by forward induction of the Arrow–Debreu prices $q_k(x)$,

$$\sum_x q_k(x)\, e^{-(\alpha_k+x)h_k} = P(0, t_{k+1}), \qquad q_{k+1}(x') = \sum_{x \to x'} q_k(x)\, e^{-(\alpha_k+x)h_k}\, p_{x \to x'}, \tag{12}$$

so that the tree reprices every discount factor of the curve exactly. Default and recovery enter as in the PDE. The backward induction applies (7) at the nodes.

3.3 Monte Carlo and Longstaff–Schwartz

The pair $(x_t, \int_0^t x_s\,\mathrm{d}s)$ is Gaussian and simulated exactly on the grid of coupon, call and (with recovery) monthly dates, which gives the state and the bank-account discount factor $D(0,t) = e^{-\int_0^t r}$ on each path without discretisation bias. The call policy is that of Longstaff and Schwartz [11], adapted to an option held by the issuer (Algorithm 1). Two choices matter:

- *Regressors.* Besides $1, z, z^2, z^3$ ($z = x/\mathrm{sd}(x_t)$), the closed-form value at t of the remaining straight flows, coupons, redemption and recovery, $S_t(x)$, computed with (4). Since the call can only lower the bond, the continuation value is below $S_t(x)$: only the paths with $S_t(x) > K + \mathrm{AI}$ can be called, and only they are regressed.
- *Out-of-sample policy.* The coefficients are estimated on 50,000 paths and applied to 50,000 independent ones. The true price is the infimum over call policies; any feasible policy gives a higher bond price, so the out-of-sample price is an *upper bound* of the true price up to Monte Carlo noise, whereas the in-sample price reuses the paths that defined the policy.

The American call is approximated by monthly call dates. Antithetic variates reduce the variance; the standard error is computed on the pair averages.

3.4 Closed forms used as benchmarks

Straight bond. With default independent of rates,

$$B_0 = \sum_i c_i P(0,T_i) Q(0,T_i) + RN \int_0^{T_n} P(0,t)\,(-\mathrm{d}Q(0,t)). \tag{13}$$

European call, zero recovery [8]. At T_c, given survival, the bond is worth $\sum_{T_i > T_c} \tilde{c}_i P(T_c, T_i; x)$ with $\tilde{c}_i = c_i Q(0,T_i)/Q(0,T_c)$, decreasing in x. With x^* such that this sum equals the strike

Algorithm 1 Longstaff–Schwartz for a callable bond (issuer's call)

1: simulate (x, D) on the dates $t_1 < \cdots < t_M$; $\Pi \leftarrow 0$ ▷ PV at 0 of the flows after t_k
2: **for** $k = M, \ldots, 1$ **do**
3: **if** t_k is a call date, strike $\kappa_k = K(t_k) + \mathrm{AI}(t_k)$ **then**
4: candidates: paths with $S_{t_k}(x) > \kappa_k$
5: fit β_k: regress $\Pi/(D(0,t_k)Q(0,t_k))$ on $\varphi(x) = (1, z, z^2, z^3, S_{t_k}(x))$ over the candidates
6: call on the candidates where $\varphi(x)^\top \beta_k > \kappa_k$: $\Pi \leftarrow \kappa_k\, D(0,t_k)\, Q(0,t_k)$
7: **end if**
8: $\Pi \leftarrow \Pi + c(t_k)\, D(0,t_k)\, Q(0,t_k) + RN\,\big(Q(0,t_{k-1}) - Q(0,t_k)\big)\,\frac{1}{2}\big(D(0,t_{k-1}) + D(0,t_k)\big)$
9: **end for**
10: re-simulate independent paths and repeat lines 2–8 with the fitted β_k; price $= \overline{\Pi}$

$\kappa = K(T_c) + \mathrm{AI}(T_c)$ and $\kappa_i = P(T_c, T_i; x^*)$, the option is a portfolio of options on zero-coupon bonds, held only if the issuer survives to T_c:

$$V_0^{\mathrm{Eur}} = B_0 - Q(0,T_c) \sum_{T_i > T_c} \tilde{c}_i\, \mathrm{ZBC}(0, T_c, T_i, \kappa_i), \tag{14}$$

where ZBC is the Hull–White zero-coupon bond call

$$\begin{aligned} \mathrm{ZBC}(0,T_c,T_i,\kappa_i) &= P(0,T_i)\Phi(h) - \kappa_i P(0,T_c)\Phi(h-\sigma_p), \\ \sigma_p = \sigma B(T_c,T_i)\sqrt{\tfrac{1-e^{-2aT_c}}{2a}}, &\qquad h = \frac{1}{\sigma_p}\ln\frac{P(0,T_i)}{\kappa_i P(0,T_c)} + \frac{\sigma_p}{2}. \end{aligned} \tag{15}$$

3.5 Spreads, yields and risk

- *OAS* [9]: the constant spread s such that discounting at $r_t + s$ (no explicit default) reprices the dirty market price with the call; the *Z-spread* is the same spread without the call. The difference measures the share of the yield that pays for the short option.
- *Yield to worst*: the lowest of the yields to each redemption date (call dates at $K + \mathrm{AI}$, and maturity at N), compounded at the coupon frequency.
- *Effective duration, convexity, DV01*: full repricing under ± 10bp parallel shifts of the zero curve, the Hull–White shift refitted to each curve, a and σ fixed. *Vega*: per bp of σ. *CS01*: per bp of parallel shift of the hazard rates.
- *Key-rate durations* [5]: triangular shifts of each pillar zero rate, linear to the neighbouring pillars, applied to the base curve sampled weekly and interpolated linearly. Bumping a single pillar of the monotone convex curve instead does not work: that interpolation is not linear in its inputs, and the single-pillar bumps add up to 7.11 instead of 7.56 for the straight bond. The triangular shifts add up to a parallel shift exactly.

4 Market Data and Calibration

Curve. The risk-free curve is the ECB euro area AAA government spot curve of 5 October 2026 (18 pillars from 3 months to 30 years; zero rates 2.85% at 1 year, 3.06% at 3 years, 3.50% at 10 years), interpolated by monotone convex [1].

Volatility. The call is an option on the co-terminal swaps, ending at the bond maturity. (a, σ) are calibrated to the ATM co-terminal European receiver swaptions $T \times (9.70 - T)$, $T = 1, \ldots, 5$, minimising the pricing errors divided by the market vega, i.e. errors in normal volatility. The normal volatilities are an *illustrative* EUR grid (no free quotes): about 77 to 79bp. Result: $a = 0.0863$, $\sigma = 111.9$bp, RMSE 0.76bp, maximum error 1.17bp.

Credit. The CDS spreads of the issuer are *illustrative* for a BBB name: 60, 85, 110, 125 and 140bp at 1, 3, 5, 7 and 10 years, $R = 40\%$. The bootstrapped hazard rates are 1.00%, 1.63%, 2.53%, 2.84% and 3.12%; the 10-year survival probability is 78.4%.

Term sheet (illustrative). Issued 15/06/2024, maturity 15/06/2036, 4.75% annual coupon, notional 100, non-call until 15/06/2029, callable at par. At the valuation date the maturity is 9.70 years, the first call 2.70 years and the accrued interest 1.4575. The European bond is callable on 15/06/2029 only, the Bermudan on the seven coupon dates 2029 to 2035, the American at any time from 15/06/2029 to maturity.

5 Validation

5.1 Closed forms

Table 1 compares the three methods with the closed forms. The PDE and the tree reprice the straight bond to machine precision (the tree by construction of α_k). The European call is recovered to 10^{-4} by the PDE and to 1.4×10^{-3} by the tree, and Monte Carlo is within 1.7 standard errors in every case.

	Case	Closed form	PDE	Tree	LSM	LSM s.e.
Straight	default-free	111.5565	3×10^{-7}	-1×10^{-12}	0.0059	0.0078
	credit, $R = 0$	92.9239	3×10^{-7}	-1×10^{-12}	0.0047	0.0062
	credit, $R = 40\%$	99.9402	2×10^{-7}	-2×10^{-7}	0.0110	0.0065
European	default-free	104.8151	-2×10^{-5}	-0.0014	-0.0066	0.0174
callable	credit, $R = 0$	92.6967	-9×10^{-5}	-0.0009	0.0090	0.0077

Table 1: Method minus closed form (dirty prices, per 100). Jamshidian's formula only holds with zero recovery.

5.2 Agreement of the three methods

Table 2 prices the four bonds with the three methods. The tree is within 0.16 cent of the PDE; Monte Carlo is 1.0 to 1.5 standard errors below it (the four estimates use the same seed, hence share their noise). For the American bond, Monte Carlo prices monthly call dates, slightly above the continuous American (Table 3). Times are 0.3 to 0.55 second for the tree and the PDE; Monte Carlo takes 6.3 seconds for the American (about 90 call dates, each with a closed-form regressor on every path).

Bond	PDE	Tree	LSM	s.e.	$\frac{\text{LSM}-\text{PDE}}{\text{s.e.}}$	PDE (s)	LSM (s)
Straight	99.9402	99.9402	99.9512	0.0065	+1.68	0.46	0.41
European	98.7382	98.7388	98.7149	0.0158	−1.47	0.46	0.49
Bermudan	98.2050	98.2034	98.1854	0.0135	−1.46	0.46	0.82
American	98.0890	98.0877	98.0758	0.0132	−0.99	0.54	6.29

Table 2: Dirty prices per 100 by method (credit, $R = 40\%$). LSM: 50,000 out-of-sample paths, monthly calls for the American.

5.3 Longstaff–Schwartz bias

Over five independent seeds, the Bermudan price is 98.1985 out-of-sample and 98.2121 in-sample, against 98.2050 for the PDE: −0.0065 and +0.0071, both within the standard error of a five-seed

mean (about 0.006). The regression policy is therefore close to optimal; the out-of-sample price is the one to report, since it values a policy the issuer could follow.

5.4 From Bermudan to American

A bond callable every $1/m$ year from the first call date must converge to the American bond as $m \to \infty$, and be worth less with more call dates. Table 3 confirms both, with the tree and the PDE agreeing at each frequency.

Call dates	annual	semi-annual	quarterly	monthly	weekly	American
PDE	98.2050	98.1491	98.1196	98.0988	98.0918	98.0890
Tree	98.2034	98.1480	98.1178	98.0970	98.0892	98.0877

Table 3: Bermudan bonds with denser call dates converge to the American bond (dirty, per 100).

5.5 Grid convergence

Figure 1 measures the error against a fine PDE (3,201 nodes, 800 steps per year), the space grid of the PDE being refined with the time grid. Crank–Nicolson with Rannacher steps converges fast for the European and Bermudan bonds (below 10^{-4} from 50 steps per year). The American converges at first order in time, as any scheme where the min is applied at every node. The tree error is not monotone (the position of the nodes relative to the strike changes with the grid), but stays below 0.2 cent from 100 steps per year. The default setting, 100 steps per year, gives errors below 0.12 cent per 100 for the three styles.

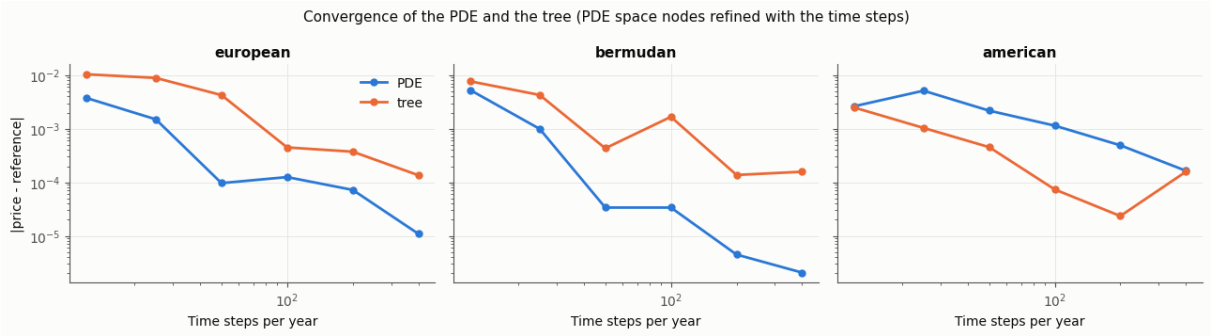


Figure 1: Absolute error against a fine PDE as a function of the time steps per year.

6 Prices, Spreads and Risk

6.1 Prices

Table 4 gives the model prices. The more call dates, the more valuable the issuer's option: it is worth 1.20 per 100 for the European call, 1.74 for the Bermudan and 1.85 for the American. The first call date carries most of the value; the Bermudan adds the right to wait, and the American only adds the dates between coupons.

Default lowers the option value by a factor of four to six: without default, the same bonds are worth 104.82, 104.29 and 104.16 against 111.56 for the straight bond, an option of 6.74 to 7.40. With credit risk, the bond trades below par, further from the call price, and the call is out of the money; moreover the issuer can only call if it survives.

	Straight	European	Bermudan	American
Dirty price	99.9402	98.7382	98.2050	98.0890
Clean price	98.4827	97.2807	96.7475	96.6314
Issuer option	–	1.2020	1.7352	1.8513
Default-free dirty price	111.5565	104.8151	104.2878	104.1586
Default-free issuer option	–	6.7415	7.2687	7.3979

Table 4: Model prices (PDE, per 100; accrued interest 1.4575).

6.2 Spreads and yields from a quoted price

In practice the clean price is observed and the spreads are solved for. Table 5 uses an illustrative quote of 98.50. The Z-spread, 139bp, ignores the call; the OAS strips it out and is lower for every style, the more so with more call dates: 106bp (European), 93bp (Bermudan) and 90bp (American). Part of the extra yield of a callable bond pays for the option the holder has sold, not for credit.

Below par, the worst case for the holder is the bond running to maturity, and the yield to worst equals the yield to maturity (4.94%). Above par it is an early call: at a clean price of 104 the yield to maturity is 4.23% but the yield to the first call is 3.17% (Figure 2).

	European	Bermudan	American
Z-spread (bp)	138.7	138.7	138.7
OAS (bp)	106.3	92.8	89.9
Yield to maturity (%)	4.941	4.941	4.941
Yield to worst (%)	4.941	4.941	4.941

Table 5: Spreads and yields at a quoted clean price of 98.50 (dirty 99.96).

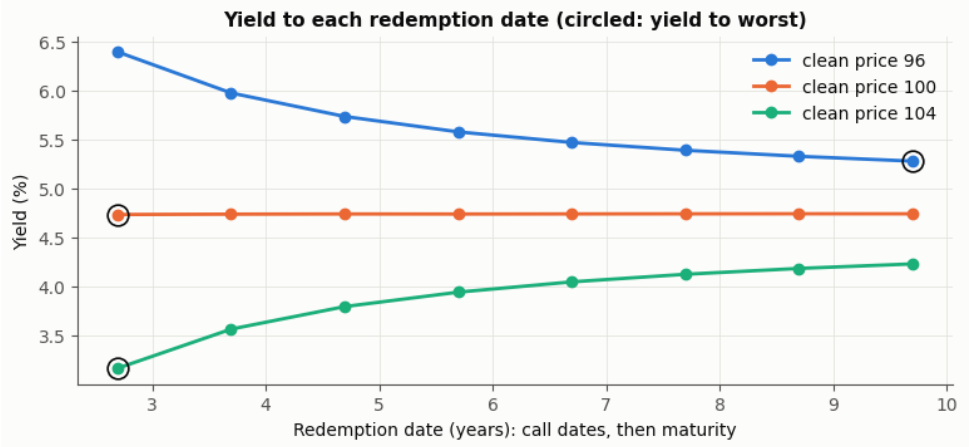


Figure 2: Yield to each redemption date of the Bermudan bond at three clean prices; circled: the yield to worst.

6.3 Interest rate, volatility and credit risk

Table 6 gives the risk measures. The callable bonds have a shorter duration than the straight bond (5.9 to 6.0 against 7.6) and a *negative* convexity: when rates fall, the call gets closer, the expected life shortens and the price is capped near the call price (Figure 3). The vega is

negative, since the holder is short the option, and grows with the number of call dates. The CS01 is smaller than that of the straight bond: a wider spread pushes the call further out of the money, which offsets part of the loss.

The three methods agree on duration, DV01, vega and CS01 to the second or third digit (e.g. Bermudan duration 5.927, 5.926 and 5.920 for the PDE, the tree and Monte Carlo, vega -0.0271, -0.0272, -0.0270). The effective convexity, a second difference, depends on the exercise discretisation: the tree (-137 for the European) and Monte Carlo (-110 for the Bermudan, policy re-estimated in each scenario) are noisy; the PDE values are the ones to use.

	Straight	**European**	**Bermudan**	**American**
Effective duration	7.560	6.044	5.927	5.894
Effective convexity	66.5	-92.0	-69.9	-70.6
DV01 (per 100)	0.0756	0.0597	0.0582	0.0578
Vega (per bp of σ)	0.0000	-0.0199	-0.0271	-0.0285
CS01 (per bp of hazard)	-0.0451	-0.0352	-0.0342	-0.0339

Table 6: Effective risk measures (PDE, ±10bp shifts, Hull–White refitted).

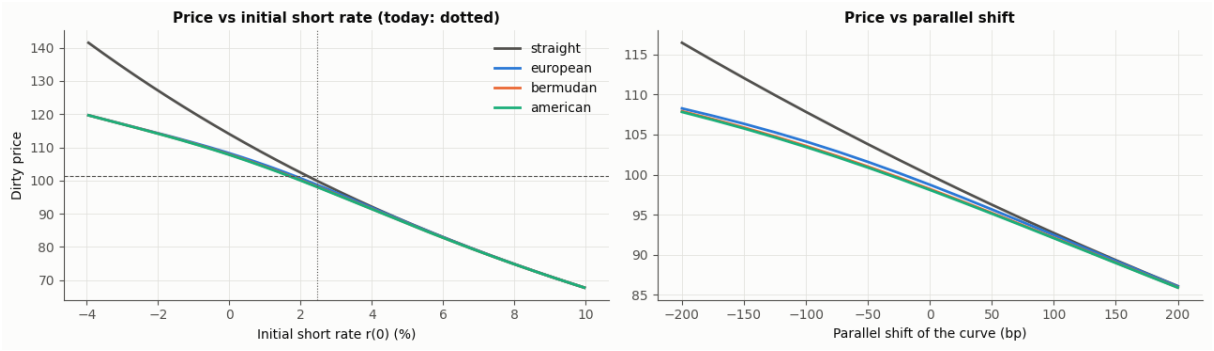


Figure 3: Left: $V(0, x)$ as a function of the initial short rate, from one PDE solve (today: dotted). Right: full repricing under parallel shifts of the curve. The callable bonds are capped near the call price when rates fall.

The key-rate durations (Figure 4) add up to the effective duration (7.5595, 6.0429, 5.9205 and 5.8996 against 7.5595, 6.0429, 5.9200 and 5.8954 for 1bp parallel shifts). The straight bond's risk sits on the 9- and 10-year pillars, near its maturity. The call moves part of it to the 2- to 4-year pillars around the first call date: a callable bond behaves partly like a bond maturing at the call. The hedge of a callable bond therefore involves short-dated instruments that a yield-to-maturity duration ignores.

7 Exercise Policy

7.1 Exercise boundary

The issuer calls when the short rate is below the critical rate $r^*(t)$ (Figure 5). On the first call date, r^* is 2.20% for the European bond, 1.54% for the Bermudan and 1.03% for the American. With more future call dates the issuer gains by waiting, and only calls for a deeper fall in rates; the European issuer, with a single chance, calls at the highest threshold. The boundary rises towards maturity (1.28% after 6.2 years, 2.44% about a month before maturity for the American; 2.55% on the last Bermudan date): with fewer coupons left to save, a smaller rate drop is enough. All thresholds are well below the forward curve (3.2% to 4.1%): the call is out

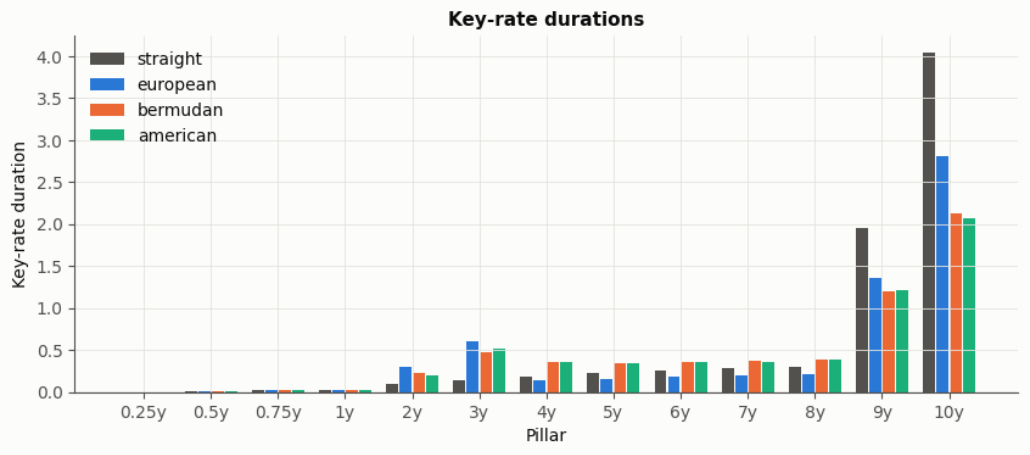


Figure 4: Key-rate durations (triangular shifts, PDE).

of the money at today's rates, because the bond, after its credit spread, is worth less than par. The tree gives the same Bermudan thresholds within one node.

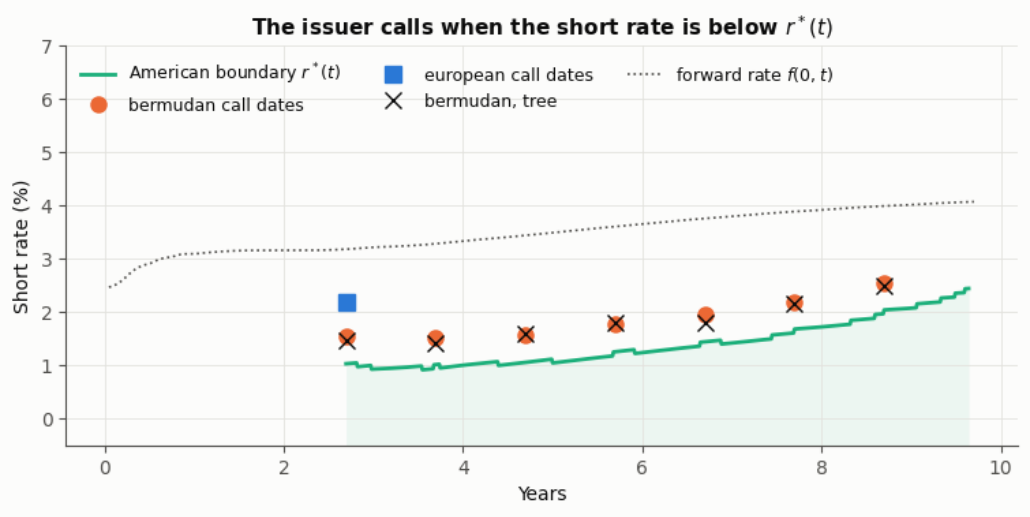


Figure 5: Critical short rate $r^*(t)$ below which the issuer calls (PDE; crosses: tree).

7.2 Call probabilities and expected life

The Monte Carlo paths give the probability of a call on each date given survival (Figure 6, Table 7). The European bond is called with probability 28%. The Bermudan bond is called with probability 43% in total, 16% on the first call date and less than 7% on each later one. The American issuer calls more often in total (46%) but spreads the calls over the call period, waiting for lower rates, so the expected life (7.60 years) is not shorter than for the Bermudan (7.59 years).

	European	**Bermudan**	**American (monthly)**
Probability of a call (%)	27.6	42.5	45.6
Expected life (years)	7.77	7.59	7.60

Table 7: Call statistics given survival (Monte Carlo, 50,000 paths); maturity 9.70 years.

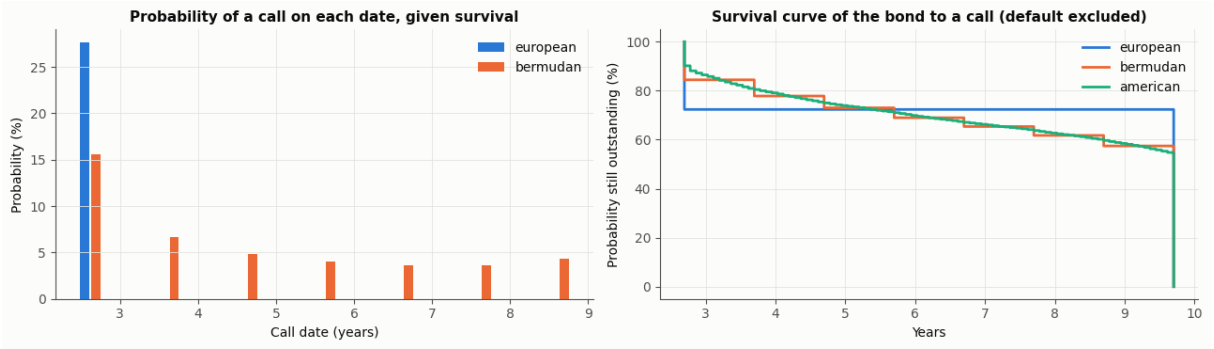


Figure 6: Left: probability of a call on each date, given survival (European and Bermudan). Right: share of the bonds still outstanding, all styles (American: monthly call dates).

8 Model Risk and Term Sheet Variants

8.1 Mean reversion

The co-terminal European swaptions fix one combination of (a, σ) for each expiry, not both parameters independently. Table 8 fixes a, recalibrates σ to the same five swaptions and reprices the bonds. The fit stays acceptable (RMSE below 1bp of normal volatility up to $a = 0.15$). The European bond, an option close to the calibration set, moves by at most 1.4 cent. The Bermudan and American bonds move by up to 0.32 and 0.44 per 100: the mean reversion sets the correlation between rates at different call dates, hence the value of choosing the call date, which European swaptions do not see. A higher a decorrelates the call dates, makes the right to choose worth more and the bond worth less. This spread is model risk at identical calibration; reducing it requires Bermudan swaption quotes or a second factor.

a	σ (bp)	**RMSE (bp)**	**European**	**Bermudan**	**American**
0.005	77.9	0.93	+0.014	+0.142	+0.182
0.020	83.5	0.88	+0.011	+0.117	+0.151
0.050	95.6	0.80	+0.005	+0.065	+0.085
0.0863	**111.9**	**0.76**	**98.7382**	**98.2050**	**98.0890**
0.150	144.8	0.91	−0.006	−0.121	−0.162
0.250	208.1	1.58	−0.007	−0.324	−0.445

Table 8: Prices with σ recalibrated for each mean reversion, as differences to the calibrated model (bold row).

8.2 Volatility and credit

The issuer's option increases with the volatility (Figure 7): from 0.01 to 0.03 per 100 at $\sigma = 30$bp to 3.07 (European), 4.21 (Bermudan) and 4.45 (American) at 200bp. It decreases with the credit spread: a parallel shift of the hazard rates of −100bp raises it to 2.54 to 3.27, +300bp lowers it to 0.05 to 0.22, as the bond moves away from the call price. With a deterministic intensity, the model ignores the other reason to call, a tightening of the issuer's own spread (Section 10).

8.3 Term sheet variants

Table 9 prices three common variants of the bond. A step-down call price (102 from 2029, 101 from 2031, par from 2033) makes early calls costly and lowers the option by 0.42 for the European bond, whose only call is at 102, and by about 0.40 for the other two. Semi-annual coupons (same

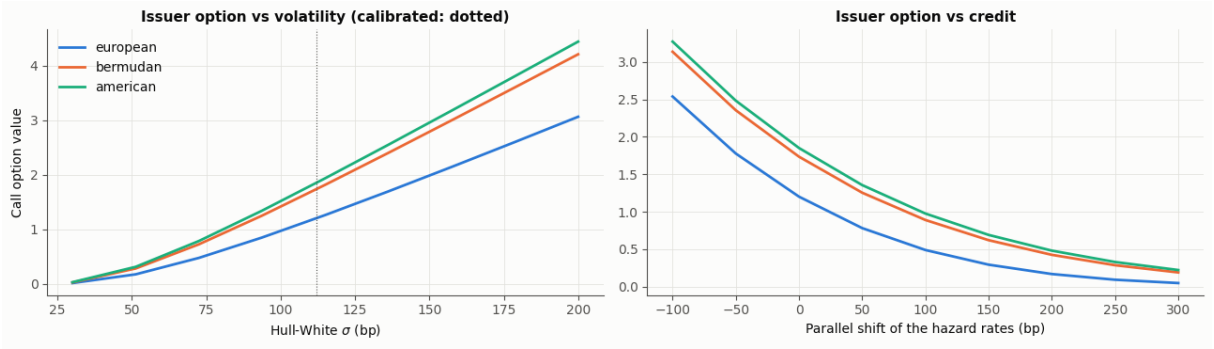


Figure 7: Issuer option value as a function of the Hull–White volatility (left) and of a parallel shift of the hazard rates (right).

annual rate) raise the bond, since coupons are received earlier, and give the Bermudan issuer twice as many call dates. Quarterly calls on the annual-coupon bond (call price plus accrued interest between coupons) lie between the annual Bermudan and the American. The tree agrees with the PDE within 0.2 cent on every variant.

Variant	Style	Dirty	Clean	Issuer option	Tree−PDE
Par call (base)	European	98.7382	97.2807	1.2020	+0.0006
	Bermudan	98.2050	96.7475	1.7352	−0.0016
	American	98.0890	96.6314	1.8513	−0.0012
Step-down 102/101/100	European	99.1624	97.7048	0.7779	−0.0014
	Bermudan	98.5970	97.1394	1.3433	−0.0012
	American	98.5088	97.0513	1.4314	−0.0012
Semi-annual coupons	European	99.1783	97.7247	1.2944	−0.0020
	Bermudan	98.5841	97.1306	1.8886	−0.0013
	American	98.5235	97.0699	1.9493	−0.0018
Quarterly calls	Bermudan	98.1196	96.6620	1.8207	−0.0017

Table 9: Term sheet variants (PDE, per 100).

9 Implementation

- `Products/callable_bond/callable_bond_pricer.py`: `CreditCurve` (CDS bootstrap), `CallableBond` and `DatedCallableBond` (term sheet), `CallableBondPDE`, `CallableBondTree`, `CallableBondLSM` (one interface: `price(bond, callable)`, `with_model`, `with_credit`), the closed forms, `option_adjusted_spread`, `effective_risk`, `yield_table`, `key_rate_durations` and `pricing_report`;
- `Fixed_Income_Models`: `YieldCurve`, `HullWhite` (zero-coupon bonds and options, exact simulation, calibration), `RatePricer`;
- `Products/callable_bond/callable_bond_pricer.ipynb`: the notebook of this report (about 4 minutes); `make_figures.py` extracts its figures.

```
from callable_bond_pricer import DatedCallableBond, CallableBondPDE,
    CallableBondTree, CallableBondLSM, pricing_report

bond = DatedCallableBond("2026-10-05", "2036-06-15", 0.0475, first_call_date="
    2029-06-15",
```

```
                        issue_date="2024-06-15", exercise="bermudan")
pde = CallableBondPDE(hw, credit, recovery=0.40)          # hw : calibrated
    HullWhite, credit : CreditCurve
print(pde.price(bond), CallableBondTree(hw, credit, 0.40).price(bond),
    CallableBondLSM(hw, credit, 0.40).price(bond))
print(pricing_report(pde, bond.with_exercise("american"), market_price=98.50))
      # OAS, YTW, duration, vega, CS01
```

10 Limitations and Next Steps

- *One factor.* Rates of all maturities are perfectly correlated; the Bermudan and American prices depend on the mean reversion at identical European calibration (Table 8). A two-factor Gaussian model (G2++) calibrated to Bermudan swaptions would reduce this model risk; the Monte Carlo pricer only needs the model's simulation and zero-coupon bond prices to extend to it.
- *Credit.* The intensity is deterministic and independent of rates, so the issuer never calls because its own spread tightens, and the option is undervalued for issuers with volatile spreads. A stochastic intensity correlated with rates (CIR or a second Gaussian factor) would capture it.
- *Market data.* Swaption volatilities and CDS spreads are illustrative; the AAA government curve is used as risk-free curve, whereas an €STR curve is the standard discounting curve for a dealer.
- *Conventions.* No business-day adjustment, settlement lag or call notice period; the American call is monthly in Monte Carlo. A notice period changes the payoff to $(K + \mathrm{AI})P(t_n, t_c; x)$ at the notice date and is a direct extension of the three methods.
- *Product scope.* Puttable bonds (holder's option, max instead of min), make-whole calls and amortising bonds are not covered.

11 Conclusion

A single pricing equation, with the issuer's min applied on the call set, prices European, Bermudan and American callable bonds under Hull–White with default risk. Three independent methods, a Crank–Nicolson PDE, a trinomial tree with unequal steps and an out-of-sample Longstaff–Schwartz Monte Carlo, agree within a fraction of a cent and reprice the closed forms. On an illustrative BBB bond, the issuer's option is worth 1.2 to 1.9 per 100, four to six times less than for the same default-free bond; it lowers the OAS by 32 to 49bp below the Z-spread, turns the convexity negative and moves the rate risk towards the first call date. The European price is pinned by the swaption calibration, but the value of choosing the call date, which separates the Bermudan and American bonds from it, depends on the mean reversion: a model risk of up to 0.44 per 100 that European swaptions alone cannot remove.

A Notation

Symbol	Meaning
$r_t = x_t + \alpha(t)$	Short rate, Ornstein–Uhlenbeck state, deterministic shift
a, σ	Mean reversion, normal volatility of the short rate
$P(t,T;x)$, $B(t,T)$	Zero-coupon bond price given the state, its sensitivity to x
$\lambda(t)$, $Q(0,t)$, R	Default intensity, survival probability, recovery rate
N, c_i, T_i, f	Notional, coupons, payment dates, coupon frequency
$K(t)$, $\mathrm{AI}(t)$, $\mathcal{T}$, T_c	Clean call price, accrued interest, call set, first call date
$V(t,x)$	Value of the callable bond before default
$x^*(t)$, $r^*(t)$	Critical state and short rate below which the issuer calls
$D(0,t)$	Bank-account discount factor $e^{-\int_0^t r}$

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